

MD SHAHFAHAD KHAN

Machine Learning Engineer | AI / ML

Mohali, India | +91-8928530995 | khannishu522@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

BCA graduate and Machine Learning Engineer with hands-on experience building end-to-end AI/ML and Generative AI applications using Python, PyTorch, Scikit-learn, FastAPI, Flask, and Docker. Skilled in Machine Learning, Deep Learning, NLP, Computer Vision, Retrieval-Augmented Generation (RAG), and Semantic Search, with a track record of deploying production-ready ML and LLM-powered solutions.

ACADEMIC PROJECTS

AI Research Assistant (RAG-based)

Jun 2026 – Jul 2026 | [Live](#) | [Github](#)

Tech Stack: Python, FastAPI, ChromaDB (Vector DB), Gemini API, PyMuPDF, Docker, Render

- Built an end-to-end AI Research Assistant using Retrieval-Augmented Generation (RAG) to enable intelligent, citation-backed question answering over uploaded PDF documents.
- Designed the RAG pipeline covering PDF parsing, text chunking, semantic embeddings, and vector similarity search using ChromaDB combined with the Gemini API for grounded answer generation.
- Developed and deployed a FastAPI application with document upload, knowledge-base management, and an end-to-end AI-powered research workflow, hosted on Railway.

DeepRank AI — Resume-Job Semantic Ranking Engine

Jun 2026 – Jul 2026 | [Live](#) | [Github](#)

Tech Stack: Python, Sentence-BERT (SBERT), Scikit-learn, Pandas, NumPy, Docker, Hugging Face Spaces

- Engineered a deep learning-based semantic resume ranking engine using Sentence-BERT (SBERT) to intelligently rank 500+ resumes against job descriptions.
- Built an AI recruitment platform featuring semantic search, weighted skill-gap analysis, duplicate resume detection, candidate recommendation, and explainable AI insights.
- Developed scalable resume processing, a recruiter analytics dashboard, and downloadable CSV/PDF/ZIP reports; deployed on Hugging Face Spaces using Docker.
- Optimized transformer-based semantic retrieval for accurate candidate ranking beyond traditional keyword matching.

Pharma Complaint AI Service

Jul 2026 – Aug 2026 | [Live](#) | [Github](#)

Tech Stack: React, FastAPI, LangGraph, Groq LLM, PostgreSQL, SQLAlchemy

- Built an AI-powered pharmaceutical complaint management system that extracts structured complaint data from **PDF, DOCX, TXT, EML, and text inputs** and automatically fills complaint forms.
- Developed a **LangGraph multi-agent workflow** for risk classification, complaint completeness, duplicate detection, summarization, root-cause analysis, and **CAPA recommendations** using Groq Llama 3.3 70B.
- Implemented a **React + FastAPI architecture** with PostgreSQL/SQLAlchemy and deployed the application using **Vercel and Render**.

TECHNICAL SKILLS

Programming: Python, SQL, C++

Machine Learning: Regression, Classification, Ensemble Learning, Feature Engineering, Hyperparameter Tuning, Model Evaluation

Deep Learning & GenAI: PyTorch, TensorFlow, CNN, Transfer Learning, Computer Vision, NLP, RAG, Semantic Search, LLM APIs (Gemini)

Frameworks & Tools: Flask, FastAPI, REST APIs, Docker, ChromaDB, Hugging Face Spaces, Railway, Render

Libraries: Scikit-learn, NumPy, Pandas, OpenCV, Matplotlib, Seaborn

Databases: MySQL, PostgreSQL, ChromaDB (Vector DB)

Dev Tools: Git, GitHub, Linux, VS Code, Jupyter Notebook

EDUCATION

Master of Computer Applications (MCA) — Specialization in AI/ML (Online)

Aug 2026 – Aug 2028

Chandigarh University, Chandigarh

Bachelor of Computer Applications (BCA)

June 2026

Chandigarh University, Chandigarh | CGPA: 7.50 / 10